Data Science Course - MiniProject III - Art Analysis

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Introduction

In many detective movies, we often see investigators constructing large boards covered with photographs, notes, and strings connecting clues to one another in order to uncover meaningful patterns or identify hidden relationships (like in Figure 1). Without this web of interconnected evidence, solving a case would be extremely challenging.

Figure 1: Visualizing complex connections to uncover hidden relationships.

A similar approach can be applied to the analysis of art history. Just as detectives connect clues, researchers can connect artists, art movements, and institutions to better understand the evolution and interdependence of artistic ideas. By examining these relationships, it becomes possible to identify patterns of influence and trace the development of art through different periods.

However, this type of analysis requires access to large amounts of digital data, making it difficult to perform using traditional manual methods such as mental associations or paper-based diagrams. To address this, I utilized a dataset from WikiArt that contains detailed information about artists, their mutual relationships, artistic movements, and educational institutions.

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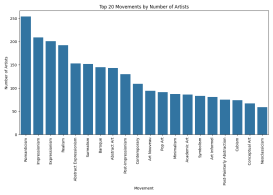
The objective of this study is to analyze these data and answer the following key questions:

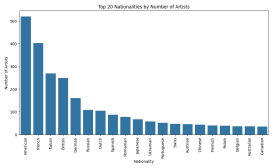
1. Which were the most influential artists?
2. Which were the most influential movements?
3. Which were the most influential institutions?
4. Which nationalities include the majority of artists?
5. Which are the largest communities in the network?

Preprocessing and Exploratory Data Analysis (EDA)

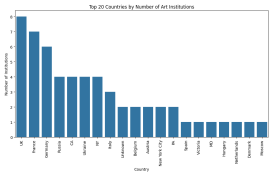
As an initial step, the dataset was examined for missing values. No records were found with a missing `url` field, which serves as the unique identifier for each entity. However, some attributes such as `nation`, `movement`, and `country` contained missing entries. These were imputed with either the value `"Unknown"` or left blank, depending on the context.



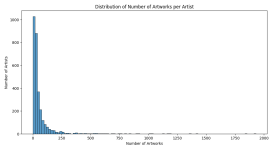
(a) Top 20 Movements by Number of Artists



(b) Top 20 Nationalities by Number of Artists



(c) Top 20 Countries by Number of Art Institutions



(d) Distribution of Number of Artworks per Artist

Figure 2: Distributions in the dataset

Subsequently, several distributions were analyzed to better understand the dataset's composition in Figure 2. These included the distribution of artistic movements, nationalities

of artists, countries of institutions, and the number of artworks per artist. The exploratory results revealed that artists from the United Kingdom, the United States, and France appear most frequently in the dataset, reflecting their historical prominence in the global art landscape.

Network Creation and Methods

Network Creation

To address the research questions, two distinct network models were constructed based on the available data.

The first network is a **directed graph** in which nodes represent artists and edges denote influence relationships between them. This network structure was primarily used to investigate the first and fifth questions, as it focuses on understanding the flow of artistic influence and the formation of communities among artists.

During initial inspection, a large number of isolated nodes were identified, as shown in Figure 3a. To improve the interpretability of the results, these isolated nodes were removed, retaining only the connected components of the network (Figure 3b).

The second network is an **undirected graph** consisting of three types of nodes: artists, movements, and institutions. Edges were established between artists and their corresponding movements or institutions, indicating the affiliations or artistic contexts in which each artist participated. This representation provides rich relational information for analyzing the second and third research questions concerning influential movements and institutions.

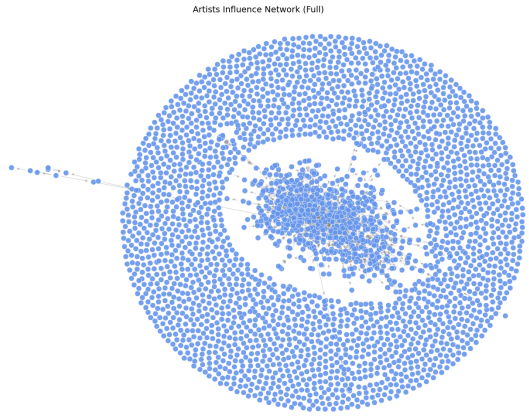
To visualize this network, a subset of ten artists was randomly selected, along with their associated movements and institutions. The resulting subgraph, shown in Figure 4, illustrates the tripartite structure of the network, where nodes are color-coded by category.

Methods and Metrics

To answer the first three research questions, it was necessary to apply suitable centrality measures to quantify the notion of influence and identify the most prominent nodes within each network. Centrality metrics allow us to determine which nodes occupy key positions in the structure—either by exerting influence over others, being connected to influential peers, or serving as important intermediaries within the network.

For the first question, concerning the most influential **artists**, the **out-degree centrality** measure was employed. This metric counts the number of directed edges originating from a node, representing how many other artists a given artist has influenced. It provides a straightforward and interpretable indication of direct influence, making it an appropriate choice for capturing the extent of each artist's impact on others within the directed influence network.

To address the second and third questions—identifying the most influential **movements** and **institutions**—the **eigenvector centrality** metric was used. Unlike degree-based measures, eigenvector centrality not only considers the number of connections a node has but also weighs the importance of those connections. In other words, a node is



(a) Full Artist Influence Network before pruning isolated nodes.

(b) Connected components of the Artist Influence Network after removing isolated nodes.

Figure 3: Comparison of the Artist Influence Network before and after filtering isolated nodes.

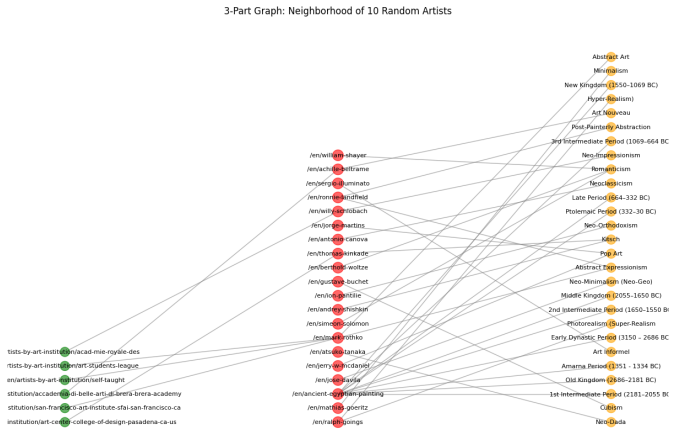


Figure 4: Excerpt of the tripartite network. Green nodes represent institutions, red nodes represent artists, and orange nodes represent movements.

considered highly central if it is connected to other highly central nodes. This makes eigenvector centrality particularly suitable for uncovering influential movements and institutions, as their prominence often arises from associations with highly regarded artists and other significant entities within the network.

Analysis and Answering the Five Questions

Question 1: Which were the most influential artists?

According to the directed influence network and the **out-degree centrality** measure, the following artists exerted the greatest influence on others:

Artist	Out-Degree Centrality (Influence Score)
Pablo Picasso	25
Paul Cézanne	23
Rembrandt	21
Caravaggio	21
Titian	17
Gustave Courbet	17
Nicolas Poussin	16
Diego Velázquez	16
Raphael	15
Henri Matisse	15
Michelangelo	15
Paul Gauguin	13
El Greco	12
Jean-François Millet	12
Wassily Kandinsky	12
Vincent van Gogh	12
Peter Paul Rubens	12
Giotto	11
Robert Delaunay	11
Giorgio de Chirico	11

Table 1: Top 20 most influential artists based on out-degree centrality. Higher values indicate a greater number of artists influenced.

These results align well with established art history, where figures like *Picasso*, *Cézanne*, and *Rembrandt* are known for shaping subsequent artistic movements and aesthetics. The dominance of Renaissance, Baroque, and Modernist masters in the list also supports the validity of this data-driven approach.

Question 2: Which were the most influential movements?

To determine the most influential artistic movements, I examined nodes of type *movement* within the tripartite network and computed their **eigenvector centrality**. The top movements were:

Movements such as *Romanticism*, *Realism*, and *Impressionism* emerged as central, reflecting their historical role as bridges between classical and modern art. Romanticism's particularly high eigenvector value suggests that it serves as a major connector among both earlier and later styles.

Movement	Eigenvector Centrality Score
Romanticism	0.5175
Realism	0.2974
Impressionism	0.2930
Academic Art	0.0950
Expressionism	0.0919
Post-Impressionism	0.0724
Symbolism	0.0687
Art Nouveau	0.0494
Orientalism	0.0448
Neoclassicism	0.0422

Table 2: Top 10 most influential art movements based on eigenvector centrality. Higher values indicate greater connectedness within the artist–movement network.

Additionally, when comparing with the movement frequency distribution obtained during EDA (Figure 2a), similar trends appear. The top movements by degree centrality were Romanticism, Impressionism, Expressionism, and Realism. While the rankings differ slightly, both analyses confirm that the 19th-century European art movements dominate the historical network, indicating a consistent structural and stylistic influence.

Question 3: Which were the most influential institutions?

Applying eigenvector centrality to institutional nodes yielded the following leading institutions:

Institution	Eigenvector Centrality
École des Beaux-Arts	0.1016
Imperial Academy of Arts (Saint Petersburg)	0.0516
Académie Julian	0.0491
Akademie der Künste	0.0331
Royal Academy of Arts	0.0319
Art Students League	0.0267
Académie des Beaux-Arts	0.0259
National Academy of Design	0.0227
Kunstakademie Düsseldorf	0.0168
Pennsylvania Academy of Fine Arts	0.0139

Table 3: Top 10 institutions by eigenvector centrality.

These institutions correspond to some of the most historically prestigious art academies, primarily located in Europe and the United States. The prominence of the *École des Beaux-Arts* and *Académie Julian* underscores the strong French influence in art education and stylistic dissemination during the 19th and early 20th centuries.

Question 4: Which nationalities include the majority of artists?

The nationality distribution of artists shows the following top 10 categories:

These results confirm that Western countries, especially the United States, France, and Italy, dominate the

Nationality	Percentage of Artists (%)
American	17.36
French	13.42
Italian	8.98
British	8.31
German	5.34
Russian	3.60
Dutch	3.50
Spanish	2.90
Romanian	2.60
Japanese	2.24

Table 4: Top 10 nationalities with the highest concentration of artists in the dataset. Percentages are relative to the total number of artists.

dataset—reflecting both their historical centrality in the global art scene and the Eurocentric bias in digitized art archives such as WikiArt.

Question 5: Which are the largest communities in the network?

Using the Louvain community detection algorithm, approximately 2,380 communities were identified in the artist influence network. The largest ones are summarized below:

Community ID	Number of Nodes
Community 461	119
Community 54	85
Community 507	62
Community 100	56
Community 655	55
Community 42	49
Community 396	40
Community 259	39
Community 761	39
Community 743	38
Community 671	34
Community 109	30
Community 197	20
Community 462	17
Community 546	13

Table 5: Largest communities detected in the artist influence network using the Louvain community detection algorithm.

Smaller communities typically represent isolated artistic circles or short-lived movements, while larger ones capture broader stylistic or temporal clusters. After filtering for communities with more than 20 nodes, twelve major clusters remained. As shown in Figure 5, these communities are well distributed and partially overlapping, indicating cross-influences between groups of artists.

Conclusion

This project explored the structure of artistic influence by constructing and analyzing networks derived from

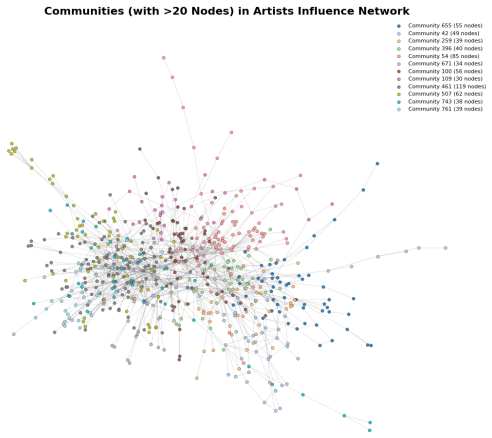


Figure 5: Communities with more than 20 nodes in the Artists Influence Network. Each color represents a distinct community detected by the Louvain algorithm. The dense central region corresponds to major interconnected art circles, while the peripheral clusters reflect more specialized or regionally isolated groups.

WikiArt data. At the beginning, it was somewhat challenging to determine how to approach the research questions—specifically, how to construct the networks and decide which features should represent nodes and edges. Eventually, I decided to build two separate graphs for this analysis; however, other configurations could also have been used, such as a single heterogeneous network including multiple node types and all relationships as edges. Another key decision involved selecting the centrality measure to define "influence." Since several different measures exist, and each captures a unique aspect of node importance, my chosen metric represents only one possible perspective. Once the general approach and methodology were established, the subsequent steps became much clearer. Overall, the results appeared reasonable and aligned with expectations. Although I anticipated that some well-known artists—such as Michelangelo—would rank slightly higher among the most influential, the overall findings make sense and provide meaningful insights into the structure of artistic influence. Overall, the study demonstrated how network analysis can yield valuable insights into cultural history. In future work, extending the dataset (to also include eastern art and culture for instance) and incorporating temporal dynamics could provide an even deeper understanding of how artistic influence evolves across time and geography.